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Impact of contraceptive use on women's health and socioeconomic status

Evidence brief

Key messages

Contraception empowers women and positively impacts their overall health and wellbeing by reducing health risks, contributing to a healthier life and conferring greater agency. Specifically, contraceptive use does the following:

- reduces health risks associated with high-risk pregnancies, such as pre-eclampsia and endometrial, ovarian and cervical cancers;
- contributes to a healthier life by improving anaemia and menstrual health, reducing the size of leiomyoma and dysfunctional bleeding and lowering pain associated with endometriosis;
- decreases depressive symptoms among women with pre-existing mental disorders and has a protective effect against anxiety in women; and
- empowers women by contributing to their socioeconomic status and wellbeing.

Health worker counselling client. © Antto - stock.adobe.com, 2024

Introduction

With an estimated 874 million women of reproductive age (15–49 years) using a modern method of contraception in 2022, *(1)* contraception use has become a norm in much of the world. Evidence has accumulated over the years regarding the multitude of medical and non-medical benefits of contraception provision, including its critical role in improving women's health by reducing unintended and high-risk pregnancies and thereby the number of women exposed to the risk of maternal death or morbidity. Contraception can potentially enhance maternal, newborn and child health by enabling women to practice the healthy timing and spacing of pregnancies and births and thus helping women to focus on their health and on caring for their newborns, infants and children. In addition to the maternal and child health benefits, the use of contraceptive methods can improve women's socioeconomic status by enabling girls and young women to continue their studies, helping them to achieve a higher education and to participate in employment that increases their earning power and contributes to their household income. As witnessed in the countries of Far East Asia and Southeast Asia, contraceptive provision also contributes to a nation's economic development by reducing fertility. This yields a demographic dividend due to the high, and increasing, ratio of working-age people compared to dependents. *(2)* In 2007, readers of the British Medical Journal (BMJ) rated oral contraceptive pills as one of the 15 biggest milestones in medicine since 1840. *(3)* The United States Centers for Disease Control (CDC) has included contraception among the 10 great public health achievements of the 20th century. *(4)* Despite this broad consensus on the benefits of family planning, some view the evidence as unconvincing primarily due to its heavy reliance on cross-sectional observational data that show associations rather than causal links or because of the varied coverage, methodologies and quality of the studies. In addition, contraception has been questioned for its demographic rationale, health rationale, quality of care and concerns for human rights. *(5)* Therefore, World Health Organization (WHO) commissioned systematic reviews of the magnitude and the quality of the evidence regarding the strength of the relationship(s) between contraceptive use and women's health outcomes as well as their socioeconomic status (SES). The overarching question for the set of six commissioned systematic reviews was to assess whether the use of contraceptives leads to improvements in women's health and their SES through the number, timing and spacing of pregnancies and to other direct health and non-health benefits. This Brief highlights the key findings and their implications.

Methods

Systematic reviews proceeded with the development of protocols and their registration on the International Prospective Register of Systematic Reviews (PROSPERO). The protocols followed the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) reporting checklist. *(6)* Relevant databases were identified, and the search strategy was guided by the Population, Intervention, Comparator, Outcome and Study Design (PICOS). Search results were exported into EndNote or Mendeley, de-duplicated and uploaded into Covidence, *(7)* a web-based systematic review software, for screening. Titles and abstracts were reviewed, and discrepancies were resolved before the review of the final set of selected publications. Data for key variables were extracted, including publication details, study characteristics, population, interventions, controls and outcomes, in a standardized data collection form. The methodological quality of studies was assessed using the updated Cochrane risks of bias tool 2 *(8)*, for randomized clinical trials [RCT]s /cluster RCTs and the ROBINS-I *(9)* for non-randomized studies of interventions, except for the reviews entitled "Does family planning use empower women? A systematic review" that used the Newcastle–Ottawa Scale and "A meta-analysis into the mediatory effects of family planning utilization on complications of pregnancy in women of reproductive age" that applied the Downs and Black scale to assess the quality.

Meta-analyses were conducted when two or more studies of similar design were available. Statistical heterogeneity and publication bias were assessed and documented. By using the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) (10) criteria, the certainty of the evidence was rated for each key outcome as high, moderate, low, or very low. The systematic review used the Newcastle–Ottawa Scale, which classifies the quality of studies into good or fair. The information on this brief refers to "women or girl" for those who were assigned female at birth. Full methodological details for each review are available in the published publications and summarized in the Annex.

Results

Use of contraception and women's empowerment

Two systematic reviews analysed the evidence on the relationship between contraception and empowerment. The first review *(11)* set out to examine the evidence on the relationship between empowerment and contraception use among adolescents (10–19 years) and young women (10–24 years). Sexual and reproductive health empowerment and agency interventions, policies and programmes were found to increase the likelihood of modern contraceptive use by 22% (Relative Risk [RR] 1.22; 95% Confidence Interval [CI] 1.02, 1.45). Additionally, multidimensional empowerment and agency interventions more than halved the incidence of adolescent pregnancy (RR= 0.48; 94% CI 0.24, 0.98).

The review identified the need for studies on the long-term impact of empowerment on contraception use and for scientifically rigorous studies on the impact of family planning on empowerment and agency among adolescent and young women.

The second review *(12)* explored the question of whether contraceptive use empowers women of reproductive age (15–49 years). Empowerment was defined as the attainment of improved health and social status by acquiring access to human development resources (such as education, skill development and self-efficacy) and to social and economic resources that contribute to attaining agency.

The systematic review found that contraceptive use empowers women. A positive impact was found on access/control of resources (education, wages or income) and on agency (decision-making and negotiation skills) in most studies. To fully understand the pathways through which family planning impacts women's empowerment, the review found a need for rigorously designed studies and the inclusion of comprehensive measures of empowerment. Ensuring access to contraceptive methods and their use among women in need facilitates attaining goals of female empowerment and agency.

Health worker taking blood pressure. © WHO / HRP / Uma Bista

Use of contraception and health risks

Four systematic reviews examined the impact of contraception on health risks.

The first review *(13)* looked at the effects of contraception utilization on pregnancy complications in women of reproductive age. The review found a substantial reduction in maternal morbidity and mortality due to pregnancy prevention and a positive impact on a range of health conditions. Women using contraception had a 40% lower chance of having high-risk pregnancies and births (95% CI 0.61, 0.80) compared to non-users. Women using low androgen hormonal contraception had a reduced risk of gestational diabetes (odds ratio [OR]= 0.84, 95% CI 0.58, 1.22). Investigating the relationship between emergency contraceptive use and ectopic pregnancy compared to non-users was not significant, with a OR of 0.93 (95% CI 0.42, 2.07) and a OR of 0.77 (95% CI 0.20, 2.93), respectively. These were not statistically significant.

Three studies showed a protective effect from the use of any modern method of contraception (Ethiopia), the use of injectables (Brazil) and the use of implants (Papua New Guinea) on maternal mortality. In Brazil, injectable DMPA (depot medroxyprogesterone acetate) use was estimated to have prevented 37–60 maternal deaths. A journal paper from the Matlab (Bangladesh) Demographic Surveillance System from 1982–2005 showed that maternal mortality fell by 60% in areas that received contraception services (OR=0.41, 95% CI 0.25, 0.67). Part of the decline was due to a fall in deaths from unsafe abortions as well as better access to emergency obstetric care and trained birth attendants. Direct obstetric mortality declined by 3% per year from 1976–1993.

The risk of maternal morbidity and mortality is reduced with the use of modern and effective long-acting reversible contraceptives (LARCs) or permanent methods primarily through the prevention of pregnancies, especially high-risk pregnancies.

The review showed a need for long-term and comparative studies investigating maternal health outcomes and any contraceptive use as well as different contraceptive methods.

The second review assessed the impact of contraceptive use on cancer risk among women of reproductive age. *(14)* The evidence from cohort studies suggests significant reductions in the risk of developing ovarian (RR= 0.66; 95% CI 0.62, 0.70), cervical (RR=0.54;95% CI 0.48, 0.62) and endometrial cancers (RR=0.69; 95% CI 0.64,0.74) among contraceptive users. No significant difference was reported between users and non-users who had breast cancer.

Compared with non-users, all hormonal contraceptive users were found to have a 36% lower risk of ovarian cancer (RR=0.64; 95% CI 0.60, 0.68). When disaggregated by type of contraceptive, the oral contraceptive pill (RR=0.62, 95% CI 0.57, 0.68) and the hormonal intrauterine device (Hormonal IUD) (RR=0.68; 95% CI 0.48, 0.96) for users of injectables (RR=1.34; 95% CI 0.81, 2.21) compared to non-users. No difference was found in the risk of ovarian cancer among non-hormonal method users (RR=0.92; 95% CI 0.80, 1.06) compared with non-users. A significant protective effect of tubal ligation was found (RR=0.66; 95% CI 0.50, 0.87) on ovarian cancer risk compared to non-users.

Contraceptive users had a significantly lower risk of endometrial cancer incidence when compared to non-users (RR=0.56; 95% CI 0.50, 0.63). All hormonal contraceptive methods had a significantly low endometrial incidence, except progestin-only pill users, which was not significant compared to non-users.

The risk for cervical cancer was higher among non-users of contraceptives compared to hormonal contraceptive users (RR=1.28; 95% CI 1.21, 1.35). When results are stratified, an increased risk was observed for oral contraceptive pill users (RR=1.43; 95% CI 1.31, 1.56) and for users of any oral and non-oral contraceptive (RR=1.19; 95% CI 1.10, 1.29), while women using a copper or hormonal IUD showed either no significant effect or a trend of lower risk on cervical cancer risk incidence (RR=0.76, 95% CI 0.52, 1.11) compared to non-users.

Tubal ligation was found to reduce the hazards of gynaecological cancer risk by 20% (HR=0.80; 95% CI 0.66, 0.87). Compared to non-users, oral contraceptive use reduces the hazard of gynaecological cancers of the ovaries and endometrium by 17% (HR 0.83; 95% CI 0.74, 0.93). The use of the oral contraceptive pill or hormonal IUD had no significant effect on breast cancer risk (RR=1.0; 95% CI 0.94, 1.06). However, BRCA1/2 mutation carriers using oral contraceptives showed a heightened risk of breast cancer (HR 1.39, 95% CI 1.15–1.67). In conclusion, the use of short-term reversible, LARC and permanent methods significantly reduces the risk of ovarian and endometrial cancer and mortality among contraceptive users compared to non-users. Future studies need to examine the mechanisms of action for each method to shed light on why, for example, DMPA may not offer the same level of protection as combined oral contraceptives. In addition, future studies should also address the gap in knowledge on the use of contraceptives by women with diseases or conditions, as most studies included only healthy women.

The third review examined the impact of contraceptive use on menstrual health among women of reproductive age *(15)*. The comparisons were made between users of all hormonal contraceptives versus users of no method or between users of oral contraceptives only versus non-users.

Two RCTs found that estradiol valerate and dienogest users were significantly less likely to experience abnormal bleeding and more likely to recover from abnormal bleeding compared to non-oral contraceptive users (RR=14.94; 95% CI 4.79, 46.65). Among women who had uterine fibroids, the use of the combined hormonal contraceptive pill reduced dysfunctional bleeding or irregular uterine bleeding compared to non-users of hormonal contraceptives. A reduction in the number of days for women experiencing bleeding and spotting was also found among contraceptive users compared to a placebo group (RR=2.17; 95% CI 0.89, 3.45). Oral contraceptive pill users compared to non-users had reduced severity of menstrual symptoms (standardized mean difference = -1.29; 95% CI -1.36, -1.03), lower pain associated with endometriosis (standardized mean difference = -0.34, 95% CI; -0.50, -0.18), less dysmenorrhea associated with endometriosis and reduction in the size of uterine leiomyoma.

Evidence from RCTs indicated that hormonal contraceptives were associated with improvements in various aspects of menstrual health, including the recovery of baseline menstrual patterns, reduced dysfunctional uterine bleeding, increased rates of amenorrhea and lower menstrual blood loss. For endometriosis management, all hormonal contraceptives showed efficacy in reducing associated pain, preventing recurrence and reducing the risk of endometriosis recurrence.

Overall, hormonal contraceptives were found to significantly reduce pain, symptom severity and abnormal bleeding patterns for women who suffer from heavy menstrual bleeding, endometriosis and uterine fibroids. However, side effects were reported, ranging from uterine and ovarian haemorrhage to headaches and abdominal pain, highlighting the need for further research into these side effects and their significance in contraceptive use.

The fourth review examined the impact of contraceptive use on mental health among women of reproductive age. *(16)* Among women with pre-existing mental disorders, both RCTs (standardized mean difference = -0.15; 95% CI -0.30, -0.00) and cohort studies (standardized mean difference = -0.26; 95% CI -0.37, -0.15) found a modest decrease in depressive symptoms among oral contraceptive users compared to non-users of any method. Hormonal contraceptive use also had protective effects for anxiety in groups with and without pre-existing mental disorders (standardized mean difference = -0.20; 95% CI -0.40, -0.01). In women without previous mental disorders, both RCTs (3 studies, SMD 0.18, 95% CI [0.02, 0.34], high quality of evidence) and cohort studies (RR 1.04 95% CI [1.03, 1.04]) detected a slight increase in the risk of depression development. In women with previous mental disorders, both RCTs (9 studies, SMD – 0.15, 95% CI [-0.30, -0.00], high quality of evidence) and cohort studies (SMD – 0.26, 95% CI [-0.37, -0.15]) detected slight protective effects of depression development. The risk of depression and of antidepressant use was, however, significantly higher among women using a hormonal IUD, implant or patch/ring compared to users of other methods. Further studies were recommended to examine the length of contraceptive use to adequately discern the impact of prolonged use on mental health. In addition, the relationship between contraceptive use and mental health needs to be studied using a life-course approach as the effects are likely to vary by age and reproductive life stage (for example, in the pre-pregnancy, pregnancy and postpartum phase). Contraceptive counselling also needs to address concerns and issues of anxiety and depression.

Summary of key findings

- Ensure the provision of quality family planning services because the evidence is compelling that access to and use of contraception improve the health and socioeconomic status of women.
- Incorporate strategies that promote women's empowerment into efforts to improve sexual, reproductive and maternal health.
- Prioritize policies, interventions and programmes that empower and strengthen the agency of adolescents and young women to prevent high-risk, unintended pregnancies. In addition, tailor policies and delivery platforms for adolescent girls and young women to their specific needs and preferences.
- Investigate the medical history of the user and conduct a physical examination. Check for symptoms of blood pressure changes, weight changes, headaches or abnormal uterine bleeding given that the health risks and benefits vary by type of contraceptive method.
- Address mental health in contraceptive counselling and method choice to provide holistic care. It also underscores the need for personalized contraceptive counselling that considers cancer risk factors.
- Strengthen health services by training healthcare providers on the risks and benefits of various contraceptive methods and by establishing well-equipped facilities and robust referral systems. Strong health systems are essential to provide quality family planning and reproductive health services and to respond efficiently to unmet needs.

Conclusions

The evidence accumulated, synthesized and assessed in WHO-commissioned reviews established the numerous health and non-health benefits of contraceptive provision and use. The primary conduit from family planning to women's improved health and empowerment is the use of modern and effective contraceptives that prevent unintended and high-risk pregnancies. Therefore, morbidity and mortality risks due to pregnancy complications and high-risk childbearing are reduced. Contraceptive use also provides young women the opportunity to avoid early and risky childbearing and enables them to realize their potential for higher education and employment leading to improved agency and empowerment in decision-making and use of resources. While confirming several health and empowerment benefits of contraceptive provision and use, these reviews also identified knowledge gaps and the need for additional scientifically rigorous studies.

Client received an injectable contraceptive. © WHO / SRH / Hickmatu Leigh

References

1. World Family Planning 2022: Meeting the changing needs for family planning: Contraceptive use by age and method. United Nations Department of Economic and Social Affairs, Population Division; 2022, (UN DESA/POP/2022/TR/NO. 4).

2. Bloom DE, Williamson JG. Demographic Transitions and Economic Miracles in Emerging Asia. World Bank Economic Review. 1998; 12:419–455. Demographic Transitions and Economic Miracles in Emerging Asia | NBER

3. Djerassi C. The Pill: Emblem of liberation. BMJ. 2007; 334:e15. (https://www.bmj.com/content/suppl/2007/01/18/334.suppl_1.DC2/milestones.pdf. (Accessed on June 26, 2025)).

4. Ten great public health achievements of the 20th century. Morbidity and Mortality Weekly Report. US: Centers for Disease Control. 1999; 48:241–3. https://www.cdc.gov/mmwr/PDF/wk/mm4812.pdf

5. Seltzer, Judith. International Family Planning Programs: Criticisms and Responses. Research Summary. Santa Monica, CA: Rand Corporation. 2002 (https://www.rand.org/pubs/research_briefs/RB5063.html).

6. Page MJ, Moher D, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, McKenzie JE. PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. BMJ. 2021 Mar 29;372:n160. https://www.bmj.com/content/372/bmj.n160. PMID: 33781993; PMCID: PMC8005925.

7. Covidence systematic review software VHI. Melbourne, Australia. [Available from: www.covidence.org.]

8. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. BMJ. 2019 Aug 28;366:l4898. doi: 10.1136/bmj.l4898. PMID: 31462531..

9. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ. 2016 Oct 12;355:i4919. doi: 10.1136/bmj.i4919. PMID: 27733354; PMCID: PMC5062054

10. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: An emerging consensus on rating quality of evidence and strength of recommendations. BMJ. 2008; 336(7650):924–26. https://www.bmj.com/content/336/7650/924

11. Lassi Zohra S, Komal Abdul Rahim, Alexandra Maria Stavropoulos, Lareesa Marie Ryan, Jyoti Tyagi, Bisi Adewale, Jaameeta Kurji, Soumyadeep Bhaumik, Salima Meherali, Moazzam Ali. Use of contraceptives, empowerment and agency of adolescent girls and young women: a systematic review and meta-analysis. BMJ Sex Reproductive Health. 2024; 0:1–17. doi:10.1136/bmjsrh-2023-202151 1.

12. Smith Nadia G Diamond, Lakshmi Gopalakrishnan, Salome Wawire, Jacqueline Kunesh, Jennifer Choi, Alison M. El Ayadi. Does family planning use empower women? A systematic scoping review. (Forthcoming)

13. Jahanfar S, Maurer O, Lapidow A, Oberoi AR, Steinfeldt M, Ali M. A meta-analysis into the mediatory effects of family planning utilization on complications of pregnancy in women of reproductive age. PLoS One. 2024 Mar 18;19(3):e0294475. PMID: 38498424; PMCID: PMC10947693. https://doi.org/10.1371/journal.pone.0294475

14. Jahanfar S, Mortazavi J, Lapidow A, Cu C, Al Abosy J, Morris K, Becerra-Mateus JC, Steinfeldt M, Maurer O, Bohang J, Andrenacci P, Badawy M and Ali M (2024) Assessing the impact of contraceptive use on reproductive cancer risk among women of reproductive age—a systematic review. Front. Glob. Womens Health 5:1487820. https://doi.org/10.3389/fgwh.2024.1487820

15. Jahanfar Shayesteh, Julie Mortazavi, Amy Lapidow, Cassandra Cu, Jude Al Abosy, Hartman C, Katherine Morris, Meredith Steinfeldt, Olivia Maurer, Bohang, Jiang, Rajkumari Anjali, Moazzam Ali. Assessing the impact of contraceptive use on menstrual health among women of reproductive age – a systematic review. The European Journal of Contraception & Reproductive Health Care, 2024; 1–31. https://doi.org/10.1080/13625187.2024.2373143

16. Jahanfar Shayesteh, Julie Mortazavi, Amy Lapidow, Cassandra Cu, Jude Al Abosy, Katherine Morris, Juan Camilo Becerra-Mateus, Paola Andrenacci, Marwa Badawy, Meredith Steinfeldt, Olivia Maurer, Bohang Jiang, Moazzam Ali. Assessing the impact of contraceptive use on mental health among women of reproductive age – a systematic review. BMC Pregnancy and Childbirth. 2024; 24:396. https://doi.org/10.1186/s12884-024-06587-9

17. WHO. Adolescent health in the South-East Asia. https://www.who.int/southeastasia/health-topics/adolescent-health

18. Wells G, Shea B, O'Connell D, Peterson J, Welch V, Losos M, et al. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses [Internet]. [Accessed June 26, 2025]. Available from: S001425651930325X:mmc1.pdf

Annex

Summary of methods applied in reviews

Use of contraceptives, empowerment and agency of adolescent girls and young women: a systematic review and meta-analysis

Databases: Cochrane Database of Systematic Reviews; Cochrane Central Register of Controlled Trials; Campbell Library; MEDLINE (PubMed); EMBASE; CINAHL and Web of Science.

Inclusion criteria:

Publication: Peer-reviewed journal publications PROSPERO (CRD42022331194).

Publication year: All available

Geographical regions: All available

Language of publication: All available

Participants: Adolescent girls (10–19) and young women (10–24) *(17)*

Interventions/exposure: Interventions that aim to increase the use of modern contraceptives or empowerment or agency. The interventions for contraceptive use included education, promotion and provision of modern contraceptives delivered through community, school and/or technological platforms, involving teachers, parents, peers and community/outreach workers. Interventions on empowerment included improving decision-making power, financial autonomy, mobility, freedom of movement, marriage/relationships, gender attitudes, beliefs/gender norms, exposure to public life, aspirations and general self-efficacy.

Comparator: No intervention/control or routine care.

Outcome: Use of any modern contraceptives (including ever use, current use); unprotected sex; and adolescent/teenage pregnancy.

Study type: Interventional studies with a control/comparison arm (randomized or non-randomized), interrupted time series studies or controlled before and after studies.

Exclusion criteria: Cross-sectional studies; commentaries, editorials, opinions, letters; conference abstracts; review articles; case reports; case studies or single intervention arm studies with no comparator.

Methodological assessment of quality:

ROBINS-I and ROB-II

Meta-analysis: Random-effects model was used to account for expected heterogeneity.

Does family planning use empower women? A systematic review

Databases: PubMed; Embase; Global Health; SCOPUS; Women's Studies International; Sociological Abstracts; Africa Wide Information; CINAHL; Global Index Medicus; Econlit; PsycINFO; Web of Science Core Collection; IBECS; PAHO; WHOLIS; IndMED; Native Health Research Database; Clinical Trials registry; WHO International Clinical Trial Registry Platform (ICTRP); and Cochrane Central Register of Controlled Trials (CENTRAL). In addition, grey literature database OpenGray/GrayNet and relevant organizational websites including the National Bureau of Economic Research (NBER); BASENet; World Bank; EngenderHealth; Eldis; 3ie; Women's group evidence; Packard Foundation; ICRW; UNFPA; Population Council; Bill and Melinda Gates Foundation were searched.

Inclusion criteria:

Publication: Peer-reviewed journal publications PROSPERO: (CRD42022318696).

Publication year: January 2000- to February 2022

Geographical regions: All available

Language of publication: English, Spanish, Swahili and Hindi

Participants: Women of reproductive age (15–49)

Interventions/exposure: Family planning use, including current use or ever use or exposure to family planning programme/intervention/policy.

Comparator: Current non-users of family planning as applicable or those not exposed to family planning programme/intervention/policy.

Outcome: A broad range of measures that have been conceptualized as empowerment were included. Publications showing a quantitative measure of the effect of family planning use on empowerment or agency were included.

Study type: Randomized controlled trials (RCTs); cluster randomized controlled trials (c-RCTs); quasi-experimental studies including regression discontinuity design, difference-in-differences, and controlled before and after (CBA) studies; pre-

post designs; longitudinal or panel studies; cohort studies and interrupted time-series studies.

Exclusion criteria:

Cross-sectional studies where temporality and directionality of the association between the use of/exposure to family planning and empowerment were not clear; publications where the family planning exposure/intervention was integrated with other topics that were related to women's empowerment, such as gender norms or gender equity; commentaries; editorials; opinions; letters; conference abstracts; review articles; case reports; case studies or single intervention arm studies with no comparator.

Methodological assessment of quality:

Newcastle–Ottawa scale *(18)*

A meta-analysis into the mediatory effects of family planning utilization on complications of pregnancy in women of reproductive age

Databases: CINAHL, Medline, EMBASE, Psycho INFO, Maternity & Infant Care, LILACS, ClinicalTrials.gov, Web of Science, SCOPUS, CENTRAL Database, WHO Latin America and the Caribbean (LILACS); WHO South-East Asia (IMSEAR); WHO Eastern Mediterranean (IMEMR); and WHO Western Pacific (WPRIM).

Inclusion criteria:

Publication: Peer-reviewed journal publications Prospero (CRD42022332783)

Publication year: January 1996 to February 2022

Geographical regions: All available

Participants: Women of reproductive age (15–49)

Interventions/exposure: Use of a contraceptive method

Comparator: No contraceptive use

Outcome: Maternal morbidity and maternal mortality

Study type: Observational studies including cross-sectional, cohort, case-control studies and any non-RCT with a comparison group.

Exclusion criteria:

Systematic reviews; scoping reviews; narrative reviews; commentaries; editorials; opinions; letters; conference abstracts and review articles.

Methodological assessment of quality:

Downs and Black Scale

Assessing the impact of contraceptive use on menstrual health among women of reproductive age – a systematic review

Databases: CINAHL, OVID Medline, Embase, PsycINFO, Maternity & Infant Care, LILACS, ClinicalTrials.gov, Web of Science, Scopus, CENTRAL Database.

Inclusion criteria:

Publication: Peer-reviewed publications Prospero(CRD42022332647).

Publication year: January 2000 to February 2022

Geographical regions: All available

Language of publication: All available

Participants: Women of reproductive age (15–49)

Interventions/exposure: Contraceptive use

Comparator: No contraceptive use

Outcome: Menstrual-related morbidity (e.g., dysmenorrhea, endometriosis, anaemia, menstrual irregularities and amenorrhea).

Study type: Interventional studies with a control/comparison arm (randomized or non-randomized), quasi-experimental, interrupted time series studies or controlled before and after studies.

Exclusion criteria:

Systematic reviews; scoping reviews; narrative reviews; meta-analyses; theses; conference proceedings; commentaries; editorials; news; protocol-only manuscripts; editorials; opinions; letters and conference abstracts.

Methodological assessment of quality:

Cochrane Database for Systematic Reviews of Interventions for RCTs and Dawns and Black for observational studies.

Assessing the impact of contraceptive use on cancer risk among women of reproductive age – a systematic review

Databases: CINAHL, OVID Medline, Embase, PsycINFO, Maternity & Infant Care, LILACS, ClinicalTrials.gov, Web of Science, Scopus and CENTRAL Database.

Inclusion criteria:

Publication: Peer-reviewed publications Prospero (CRD42022332647).

Publication year: January 2000- to February 2022

Geographical regions: All available

Language of publication: All available

Participants: Women of reproductive age (15–49)

Interventions/exposure: Contraceptive method use

Comparator: No contraceptive use.

Outcome: Mortality and morbidity due to reproductive tract cancers (e.g., ovarian cancer, endometrial cancer, cervical cancer and breast cancer).

Study type: Cluster randomized controlled trials, controlled clinical trials, controlled before and after studies, interrupted time series studies, cohort or longitudinal analyses, regression discontinuity designs and case-control studies.

Exclusion criteria:

Systematic reviews; scoping reviews; narrative reviews; meta-analyses; theses; conference proceedings; commentaries; editorials; news; protocol-only manuscripts; editorials; opinions; letters; conference abstracts and studies with a loss of follow-up exceeding 20%.

Methodological assessment of quality:

Cochrane Database for Systematic Reviews of Interventions for RCTs and Dawns and Black for observational studies.

Assessing the impact of contraceptive use on mental health among women of reproductive age – a systematic review

Databases: CINAHL, OVID Medline, Embase, PsycINFO, Maternity & Infant Care, LILACS, ClinicalTrials.gov, Web of Science, Scopus, CENTRAL Database, WHO Latin America and the Caribbean (LILACS); WHO South-East Asia (IMSEAR); WHO Eastern Mediterranean (IMEMR); and WHO Western Pacific (WPRIM).

Inclusion criteria:

Publication: Peer-reviewed journal publications Prospero (CRD42022332647).

Publication year: January 2000 to February 2022

Geographical regions: All available

Language of publication: All available

Participants: Women of reproductive age (15–49)

Interventions/exposure: Modern contraceptive method use

Comparator: No contraceptive use.

Outcome: Any effect on mental health status, including mood disorders such as depression, bipolar and anxiety disorders and psychotic disorders such as schizophrenia and post-traumatic stress disorder (PTSD).

Study type: Parallel or cluster randomized controlled trials, controlled clinical trials, controlled before and after studies, interrupted time series studies, cohort or longitudinal analyses, regression discontinuity designs and case-control studies.

Exclusion criteria:

Systematic reviews; scoping reviews; narrative reviews; meta-analyses; theses; conference proceedings; commentaries; editorials; news; protocol-only manuscripts; editorials; opinions; letters; conference abstracts or studies with no comparator.

Methodological assessment of quality:

Cochrane Database for Systematic Reviews of Interventions for RCTs i.e. Cochrane RoB-2 and Dawns and Black for observational studies.

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